ECON 300 – Labour Economics

Practice Problem Set 5

Question 1. A country admits 320,000 immigrants in a year. Of these:

- 160,000 are economic immigrants,
- 96,000 are family-class immigrants,
- 64,000 are refugees or other humanitarian immigrants.

Answer the following:

(a) What percentage of total immigrants are economic immigrants?

(b) What percentage are family-class immigrants?

(c) What percentage are refugees or humanitarian immigrants?

(d) If next year the number of economic immigrants rises by 20% while the other two categories remain unchanged, what will be the new total number of immigrants?

(e) Under the new total, what share of all immigrants will be economic immigrants?

Question 2. In a local labour market, the annual labour demand and labour supply for low-skilled labour are given by:

$$W = 32 - 0.02N^D$$

$$W = 8 + 0.01N^S$$

where W is the hourly wage and N is the number of workers.

(a) Find the initial equilibrium wage and employment.

(b) Suppose immigration increases labour supply by 120 workers at every wage, so the new supply curve becomes:

$$W = 8 + 0.01(N^S - 120)$$

Find the new equilibrium wage and employment.

(c) Calculate the change in wage and the change in employment caused by immigration.

(d) Briefly interpret the result.

Question 3. Suppose immigrants and native-born workers are imperfect substitutes. The labour demand for native-born workers is:

$$W = 40 - 0.01N_n + 0.005N_i$$

where:

- W = native-born wage,
- N_n = number of native-born workers,
- N_i = number of immigrant workers.

Initially, $N_i = 200$, and native-born labour supply is:

$$W = 10 + 0.02N_n$$

(a) Find the equilibrium wage and employment of native-born workers when $N_i = 200$.

(b) Suppose immigration rises so that $N_i = 300$. Find the new equilibrium wage and employment of native-born workers.

(c) Explain whether immigration helps or hurts native-born workers in this example.

Question 4. A 25-year-old worker compares the present value of lifetime earnings in Country A and Country B. The worker expects to work for 35 more years.

- Annual earnings in Country A: \$52,000
- Annual earnings in Country B: \$58,000
- Annual non-monetary migration cost equivalent: \$2,000
- One-time moving cost: \$15,000
- Discount rate: 4%

Assume annual earnings are constant over time.

(a) Compute the annual net earnings advantage of Country B over Country A.

(b) Compute the present value of this annual earnings advantage over 35 years using the annuity formula:

$$PV = A\left(\frac{1-(1+r)^{-T}}{r}\right)$$

(c) Subtract the one-time moving cost and determine whether migration is worthwhile.

(d) If the moving cost rises to \$30,000, does the decision change?

Question 5. Consider the following yearly earnings profile of an immigrant relative to a comparable native-born worker:

Years Since Arrival	Immigrant Earnings ($)	Native-born Earnings ($)
0	32,000	44,000
5	39,000	47,000
10	46,000	50,000
15	52,000	54,000

(a) Compute the earnings gap at arrival.

(b) Compute the earnings gap after 5, 10, and 15 years.

(c) By how much has the immigrant closed the earnings gap after 15 years?

(d) What percentage of the initial gap has been closed after 15 years?

Question 6. In a point system, applicants are scored as follows:

- Education: up to 25 points
- Official language ability: up to 28 points
- Work experience: up to 15 points
- Age: up to 10 points
- Arranged employment: up to 10 points
- Adaptability: up to 10 points

An applicant receives:

- Education: 21
- Language: 20
- Work experience: 11
- Age: 8
- Arranged employment: 0
- Adaptability: 7

(a) Calculate the total points.

(b) If the passing score is 67, does the applicant qualify?

(c) If the applicant improves language points by 5, what is the new total?

(d) What is the minimum number of additional points needed to reach 67 if the applicant initially fails?

Question 7. In a labour market:

- Total working-age population = 2,400,000
- Employed = 1,560,000
- Unemployed = 120,000

(a) Calculate the labour force.

(b) Calculate the unemployment rate.

(c) Calculate the labour force participation rate.

(d) Calculate the employment rate.

Question 8. Suppose a country has:

- Labour force = 1,800,000
- Unemployment rate = 7.5%

(a) Calculate the number of unemployed workers.

(b) Calculate the number of employed workers.

(c) If employment rises by 45,000 and unemployment falls by the same amount, what is the new unemployment rate?

Question 9. In a given month, the following labour market transitions occur:

- 210,000 unemployed workers find jobs
- 180,000 employed workers lose or leave jobs and begin searching

(a) Calculate the net monthly change in unemployment.

(b) If the initial stock of unemployed workers is 960,000, what is the stock of unemployed workers at the end of the month?

(c) If the labour force is 16,000,000, what is the unemployment rate at the end of the month?

Question 10. In a labour market, the unemployment rate can be approximated by:

$$UR = \text{Incidence} \times \text{Duration}$$

Suppose:

- Region A has incidence = 2.5% per month and average duration = 3 months
- Region B has incidence = 1.5% per month and average duration = 5 months

(a) Calculate the unemployment rate for Region A.

(b) Calculate the unemployment rate for Region B.

(c) Which region has the higher unemployment rate?

(d) Explain in words how the same unemployment rate could arise from different combinations of incidence and duration.

Question 11. The following information is given for two age groups:

Group	Unemployment Rate	Incidence Rate	Duration (months)
Youth (15–24)	12.0%	6.0%	2.0
Adults (25–54)	5.6%	2.0%	2.8

(a) Verify approximately whether $UR \approx \text{Incidence} \times \text{Duration}$ holds for each group.

(b) Which group has higher unemployment incidence?

(c) Which group has longer unemployment duration?

(d) Explain why youth unemployment rates can be high even when average duration is short.

Question 12. A labour force survey reports:

- Working-age population = 5,000,000
- Labour force participation rate = 66%
- Employment rate = 61%

(a) Calculate the labour force.

(b) Calculate the number employed.

(c) Calculate the number unemployed.

(d) Calculate the unemployment rate.

Question 13. In a recession, the following occurs:

- 40,000 workers become discouraged and leave the labour force
- 30,000 workers lose jobs and start actively searching
- Initially:
  - Labour force = 3,000,000
  - Unemployed = 180,000

(a) What is the initial unemployment rate?

(b) What is the new number of unemployed workers?

(c) What is the new labour force?

(d) What is the new unemployment rate?

(e) Briefly explain why the unemployment rate may not fully capture hardship when discouraged workers leave the labour force.

Question 14. Consider the following long-term unemployment data:

Country	Total Unemployment Rate	Share Unemployed 12+ Months
Country X	6.0%	10%
Country Y	6.0%	35%

(a) Calculate the long-term unemployment rate for Country X.

(b) Calculate the long-term unemployment rate for Country Y.

(c) Which country has the more serious long-term unemployment problem?

(d) Explain why two countries with the same unemployment rate may have very different labour market conditions.

Question 15. Suppose Canada has:

- Unemployment rate = 5.8%
- Labour force participation rate = 65.5%
- Working-age population = 31,000,000

(a) Calculate the labour force.

(b) Calculate the number of unemployed workers.

(c) Calculate the number of employed workers.

(d) Calculate the employment rate.